AI DEVELOPMENT FOR PHOTOVOLTAIC INSTALLATION FORECASTING

Guillermo Blanco[2] [0009-0003-1013-6892], Marco Antonio Melgarejo[1] [0009-0004-2358-6953], Alejandro Pérez, PhD.[1] [0000-0002-4262-9275], and Camilo Palazuelos, PhD.[2] [0000-0003-4132-9550]

[1] Centro Tecnológico CTC, Parque Científico y Tecnológico de Cantabria, 39011 Santander, Spain
[2] Universidad de Cantabria, Av. de los Castros s/n, 39005 Santander, Spain

Abstract. The photovoltaic industry is one of the mains inside the field of renewable energy, with Spain being one of its headliners. Therefore, the planification of the photovoltaic plant production is key to administrate the energetic plan of the beneficiary.
This project will detail the development of a prediction model, trained through the exploit of data coming from a pilot photovoltaic plant located in Santander, Spain. In base of this model, it will be possible to know the expected generated power and administrate the energetic charges in an optimum way.
The content will focus both on the tools used in the development and the project's life cycle, exposing the difficulties found through the process and how they were solved, resulting in a prediction model with a R^2 of 0.92.

Keywords: Machine Learning · Photovoltaic · Artificial Intelligence.

1 Introduction

The spanish photovoltaic market is divided between productors and self-consumers. Productors are those who sell their energy to the electrical network. On the other hand, self-consumers do not sell their production, instead, they can save it for themselves. In Spain, a regulatory barrier has been implemented, restricting photovoltaic installations. Some self-consumers, normally the ones with a production of over 100kW, are forced to comply with the zero-injection requirement, ensuring no power is fed into the grid.

The zero-injection requirement is a law which prohibits the injection of the production into the electrical network. Therefore, the self-consumers must adjust the photovoltaic production in order not to exceed the electrical demand of the installation [1].

The energetical flexibility in this context defines the capacity of adaptation of the demand depending on external agents such as rates or impositions. It has been proved that it is possible to save around 15-25% of the total spend when a predictive control of the production is being used, while it also contributes to lower the CO_2 emission. In 2019, it was believed that photovoltaic had prevented 7 million of tons of CO_2 to be casted to the atmosphere, amount equivalent to 3 million cars circulating throughout a year [2] [3].

Optimization is divided in two types, according to its duration. The short-term optimization focus on hourly energetical production to minimize costs and increase efficiency in a short period of time. The long-term optimization tries to determine the optimal dimensions of the installation and components while minimizing the cost of installation. In order to calculate the electrical demand of an installation, we need to consider multiple factors depending on time, as shown in eq. (1):

$$D(t) = B(t) + P(t) + N(t) + A \tag{1}$$

Where $B(t)$ represents the basic work cycle, $P(t)$ the peak demands, which introduce the maximum demand events without overlapping, $N(t)$ the noise, which measures the stochastic noise fluctuations. Last, A means the base consumption, assuring realistic conditions [4].

Spain has gone from 11GW of photovoltaic installed power in 2020 to more than 32GW by the end of 2024 [5], becoming the main production technology, even surpassing the eolic. Nowadays, photovoltaic energy means the 17% of the total national energy, making Spain one of the headliners in solar energy in Europe.

Internationally, photovoltaic represents 46% of renewable energy, and the International Energy Agency (IEA) estimates around 3.6TW of new renewable energy before 2030, being 80% of this photovoltaic. As 2030 gets closer, photovoltaic industry will increase its presence as the main renewable type of energy.

Comparing to other renewable energies, photovoltaic gets much more positive results in environmental impact. Polluting during its manufacturing process are lower than other types of energy, such as eolic.

In spite of having accomplished this achievements, photovoltaic industry still has some challenges, being the prize of the energy one of the main. Being a tecnological sector in such a big proliferation, internal competence has raised significantly, which has caused a fall in the sell prize of solar energy. This situation has directly affected the installations profitability [6]. If harsh actions are not taken soon, the situation is expected to worsen by the end of 2026.

The goal of this project is to create a prediction model that, mainly, helps installations under the zero inject rule to administrate their demand through forecasts, but it will also be helpful for any kind of intelligent management system of energy control. It must predict the expected value of the power generated by the solar installation.

2 Related work

Photovoltaic power forecasting has gained prominence amid Spain's rapid solar expansion. Early models employed statistical data but limited by non-linear weather dependencies.

Machine learning advancements shifted toward tree-based ensembles. *Random forests* and *Extra Trees* improved handling of meteorological inputs like irradiance and temperature, achieving robust short-term predictions on UCI datasets. *Gradient boosting*, particularly *XGBoost*, outperformed these via optimized regularization and feature importance, with R^2 scores exceeding 0.90 in benchmarks [7].

As shown in Table 1, in Spain tools such as Shirenda [8] leverage machine learning on REE data (2015–2020) for nationwide yield estimation, supporting zero-injection compliance under RD 244/2019. Recent studies forecast rooftop PV with tuned *XGBoost* models and emphasize real-time AEMET integration, but they still overlook edge deployment [9].

These contributions enable demand optimization, saving 15-25% costs, but gaps persist in low-resource hardware forecasting and hybrid *skforecast* [10] with *XGBoost* for real pilot data [11], motivating our methodology.

Table 1. Comparison of related forecasting studies

Work	**Method**	**Horizon**	**Key Metric**	**Limitation**
Chen (2016) [9]	*XGBoost*	Short-term	R^2: 0.92	Offline focus
Zhang (2023) [12]	*LSTM-AdaBoost*	Short	RMSE: 0.043	Synthetic data
Shirenda_PV (2025) [8]	ML on REE	Long-term	N/A (yield est.)	No real-time

This project will further explore the application of machine learning models, especially *XGBoost*, in the renewable energy domain. In particular, it will analyze how these models can capture non-linear relationships between meteorological variables and photovoltaic generation, improving forecasting accuracy and supporting smarter operational decisions for solar installations.

3 Methodology

3.1 Data source

The data was obtained from a real photovoltaic installation on the roof of CTC, throughout a whole year. It was necessary to put together the dataset of power generated and the meteorological dataset, and sincronize them temporarily. This was repeated for another photovoltaic installation in Lorca, but on this paper we will focus on the one in Santander.

3.2 Dataset preprocessing

When trying to create a prediction model, the first step to take is to make sure that the dataset which will be used to train it is prepared and processed. This must have similar conditions to the place where we try to predict, in order for the model to create patterns of that place in particular. This have a direct impact in the accuracy of the prediction.

Once we have the final dataset, an analysis is needed, in order to find and eliminate possible outliers which would harm the precision of the model.

Percentile25 is the value below which is the 25% of the data. Percentile75 is the same, with 75% instead. The interquartile range (IQR) is the difference between them both [13]. Removing the outliers causes a not too large reduction of the dataset, which is favorable, but it will also leave gaps throughout the dataset.

There are some samples with a value of 0 that go against logic, probably due to bugs in the register system. We need to remove them, because they would increase the prediction error falsely. The way used to correct them was through linear interpolation.
Next Figures illustrate the effect of linear interpolation on the time series: Figure 1 shows the original signal with anomalous zero-valued samples, while Figure 2 shows the corrected signal after replacing those points through interpolation.

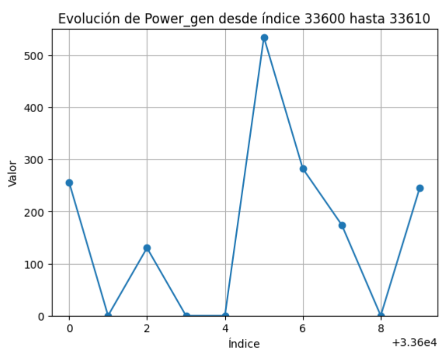

Fig. 1. Before interpolation

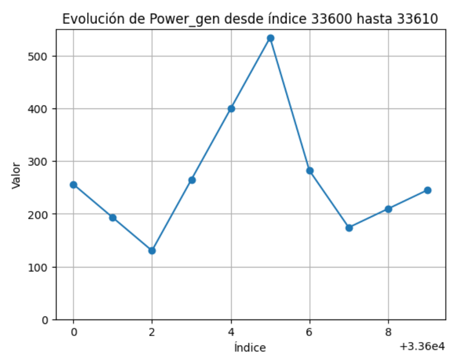

Fig. 2. After interpolation

Finally, the dataset had some of its rows with a NaN value in one or more columns, so the final step was to remove the rows where this happened. After having done all the processing, this is the final dataset we obtained:

Table 2. Dataset descriptive statistics

Statistic	Power_gen(W)	temperature(C^{o})	windSpeed(km/h)	solarRad(W/m^{2})	moist(%)	rain(mm)	pressure(hPa)
count	41821.00	41821.00	41821.00	41821.00	41821.00	41821.00	41821.00
mean	86.63	16.32	6.87	101.45	79.29	0.53	30.13
std	156.52	4.22	5.99	162.07	10.71	1.24	0.20
min	0.00	4.44	0.00	0.00	46.00	0.00	29.60
25%	0.00	12.90	3.22	0.00	73.00	0.00	30.00
50%	0.00	16.30	4.83	0.00	82.00	0.00	30.20
75%	94.00	19.80	9.66	153.00	88.00	0.00	30.30
max	780.00	30.60	27.40	631.00	96.00	5.40	30.70

Due to the fact that the variables of the dataset are scaled differently, is a good practice to scale them. Its crucial to perform this after spliting into train and test data, because, if doing so before, we would be leaking information from the training data into the test data. This practice is known as data leaking, and it decreases the performance of the model.

Scaling makes all variables go within the same range, commonly between 0 and 1. To achieve this we will use the *MinMaxScaler*.

After the processing, we can begin the developing and training process of the model.

3.3 Model development

Temporal series are data ordered chronologically. Forecasting consists on predicting the future value of a temporal serie, being able to use its past behaviour autoregresively using *skforecast* and external variables [11]. We will use exogenous variables such as solar irradiance, temperature or wind speed, as well as endogenous variables, like the mean, median, maximum or minimum value. Temporal series differ from classic regression model because of the fact that the pass of time plays a starring role, but, in the end, they are regression models, like *Gradient Boosting* or *Random Forest*.

A forecaster model was trained using a *XGBoost*. In order to calculate the optimum parameters of the *XGBoost* a *GridSearchCV* was used. After the training, we could see the importance of each variable and which one contributed more to the result of the model:

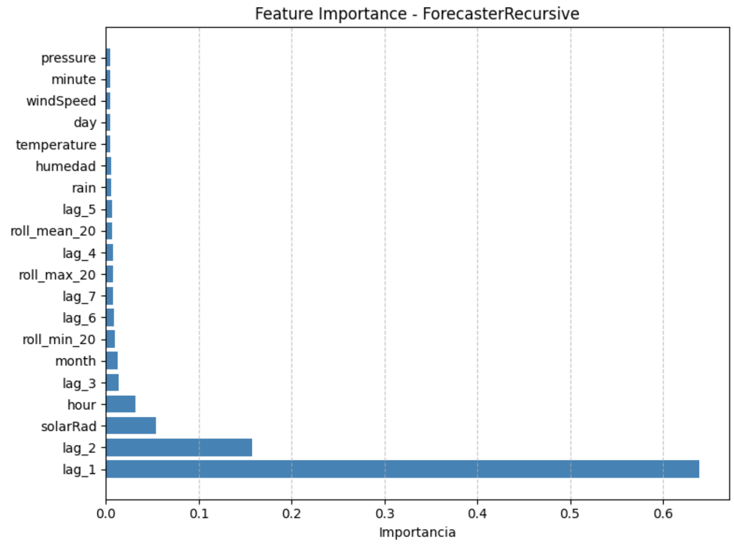

Fig. 3. Feature importance

It is clear that the most important variables are the samples which precedes the predicted value, which is normal, due to the fact that this variables have the value of the objective variable 5 and 10 minutes earlier. The rest of the variables have little importance, being the solar irradiance and the hour of the day the ones following the lags. This makes sense physically. The minimum and maximum value of the 20 previous samples makes it also into the top 10. Although it would seem that we could only use the Lag_1 and Lag_2 variables, this would be an error because, even though the rest of the variables have little impact on the result, they help to reduce the error in prediction.

It is vital to make sure that we are not overfitting our model. This means not making the model memorize the training data instead of learning prediction patterns. This would result in great metrics when using the same data to test, but a complete failure once we test the model with extern data. One way to prove the non-existance of overfitting is observing how the error in validation and training data progresses through iterations. If the error increases considerably in the validation but continues to fall in the training, we have an overfitted model. We can see that we are not in front of an overfitted model:

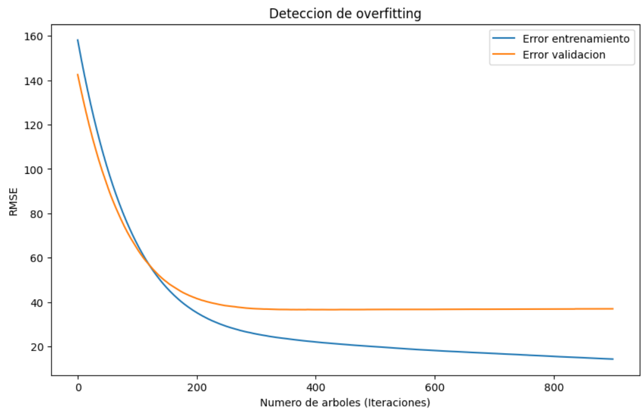

Fig. 4. Training and validation error across iterations.

4 Results

4.1 Prediction metrics

After the prediction of the test data, we got the next results:

Table 3. Prediction metrics

MAE	16.5215 (W)
MSE	1631.2590 (W)
RMSE	40.3888 (W)
R^2	0.9212

The results are not perfect but prove that the model is robust and covers a good part of the variable variance.

4.2 Graphical representation

A good way to prove the accuracy of the model is to put next to each other the real values and the predicted ones. The figure below represents in red the predicted values, and in blue the actual value of the test data. In general, this was the whole test data:

It is interesting to prove the accurracy at a lower level, trying to predict the objetive variable in the next 24 or 48 hours. This works as the figure before, red for prediction and blue for real value. Below are the predictions for 24 and 48 hours ahead, what could be the most useful information that the installation may need. We can see that they have a very similar form, proving that the model is capable of predicting the tendency of the power generated by the installation:

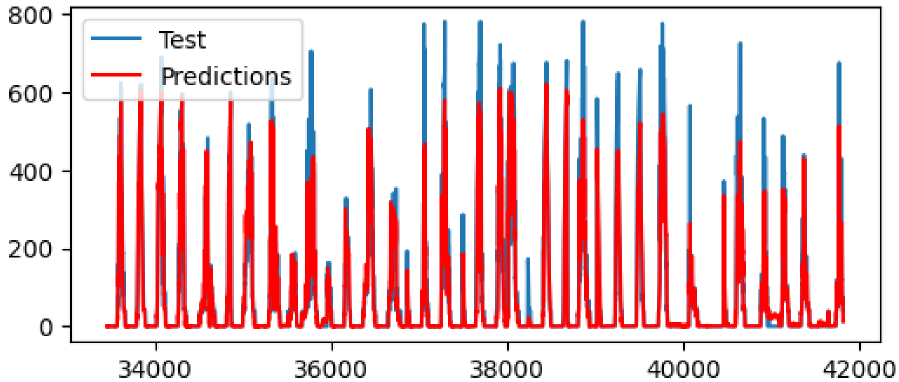

Fig. 5. Prediction through the whole dataset.

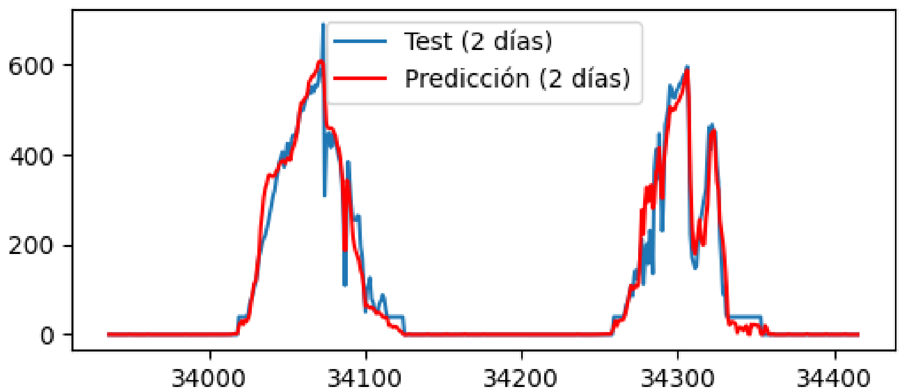

Fig. 6. Prediction of the next 48 hours.

Another way to visualize how the prediction went is not through time but putting all the data together. The more it looks like the discontinuous red line, the better. On the axis Y is the predicted value, and on the axis X the real value:

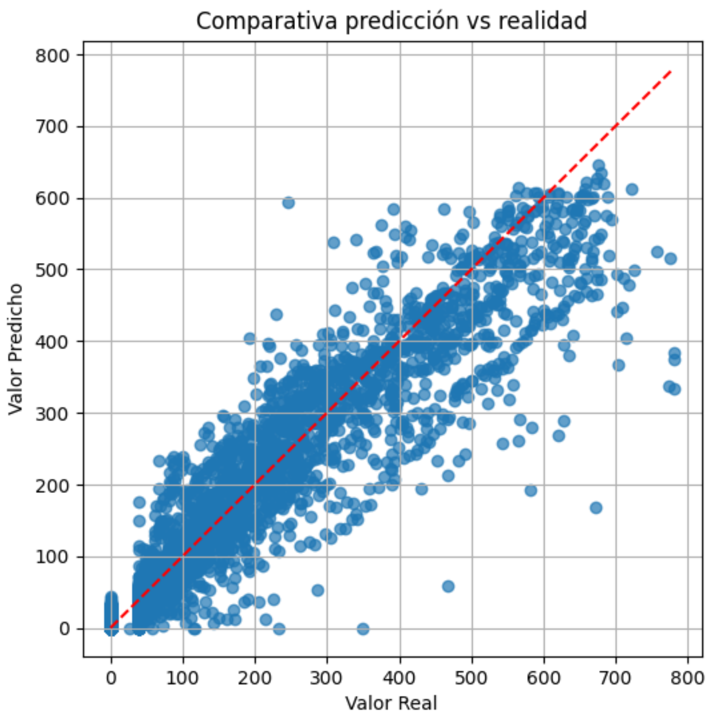

Fig. 7. Predicted vs. real values for the full test set.

4.3 Practical applications

This model will be helpful in those cases where a highly demandant energetic process has to take place, because it allows the operator to know when it should be done, whether it has to be done now because a fall in production is expected, or the other way around, if it wold be better to wait until an increasement happens. For example, it could be useful to know if it will be more production in the morning or in the afternoon. This way, it is easier to administrate the charges throughout the whole week. On the following figure, the morning average is in light blue, the afternoon average in yellow and the daily average in green:

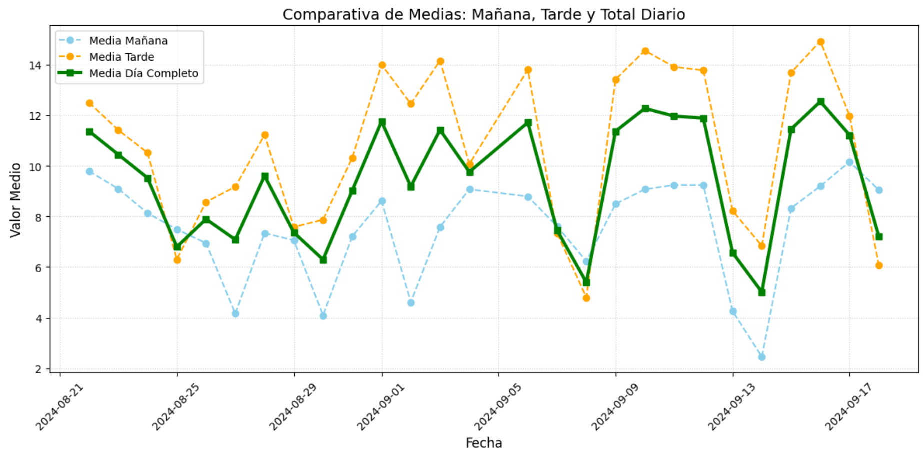

Fig. 8. Average production profile for the Lorca installation.

5 Conclusions and future steps

The development of a prediction model does not make sense whitout a practical purpose. As was mentioned earlier, the objective of this project was to handle the electrical control of an installation under the zero inject rule. Thanks to the model, it is possible to expect how much energy you are going to produce, and adjust the electrical demand in consequence.

Compared to the state of the art, the developed methodology demonstrates highly competitive performance while addressing key literature gaps. The model achieved an R^2 score of 0.9212, which matches the robust short-term predictions of previous benchmarks, such as the 0.92 score achieved by offline-focused models. More importantly, while other recent studies often rely on synthetic data or lack real-time edge deployment capabilities, this approach successfully utilizes hybrid skforecast with XGBoost on real pilot data. This directly overcomes previous limitations and lays the groundwork for low-resource hardware forecasting.

In the future, it would be interesting to syncronize the model to AEMET in real time. AEMET is the entity which provides the meteorological data daily in Spain. Doing so would allow the model to work independently, and not need someone to manually tell it the exogenous variables.

Also, implementing a graphical interface would ease the process of prediction to anyone not familiarized with this kind of models. For example, designing a UI that shows you the prediction for the rest of the day and the following day, and allows you to view the prediction for a day in the future in particular..

Disclosure of Interests. the authors have no competing interests to declare that are relevant to the content of this article.

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